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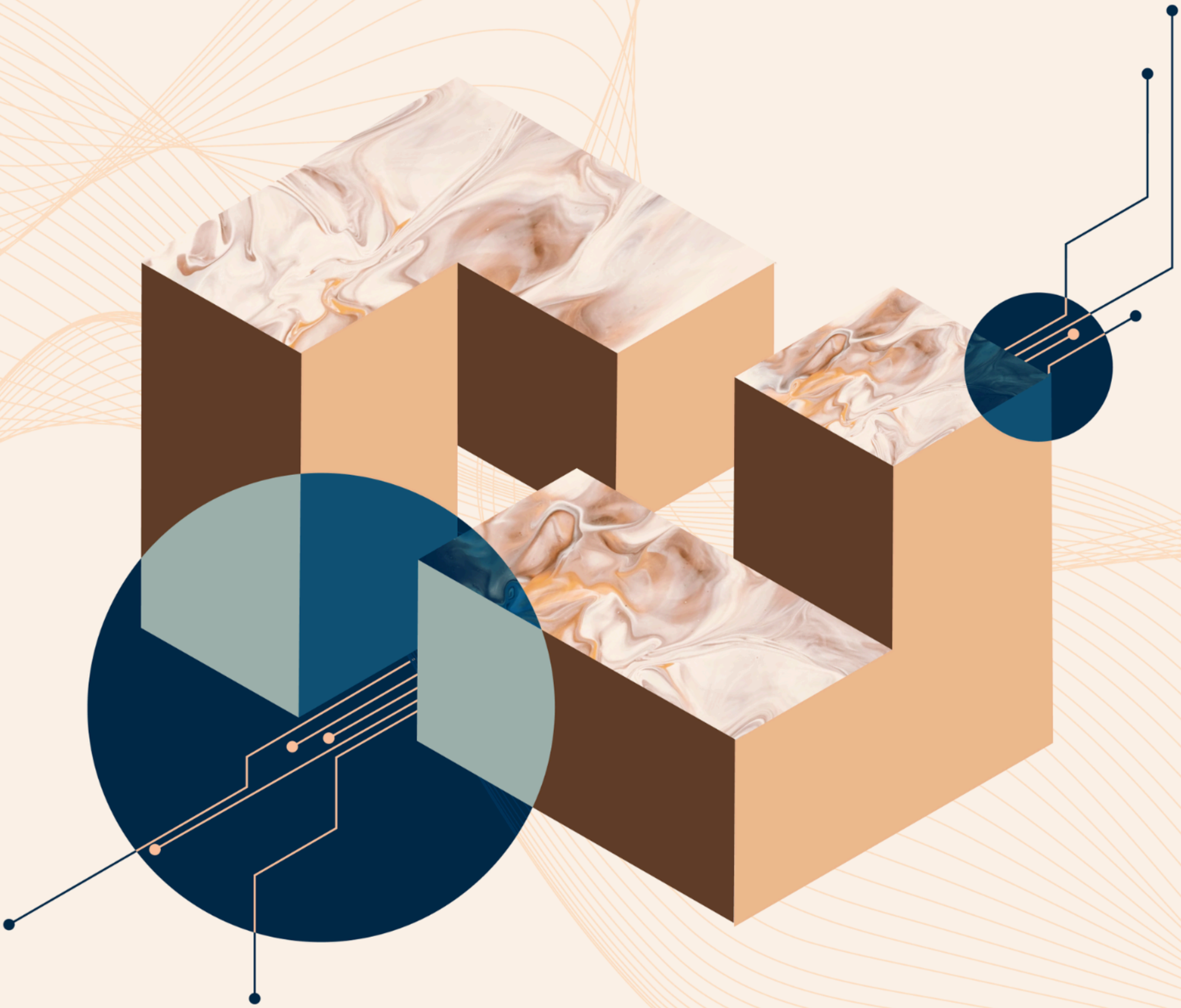

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Strategic Visions in AI Governance: Mapping Pathways to Victory

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Executive Summary

What AI policy objectives should we work towards? This depends greatly on one's *strategic vision*. Strategic visions are high-level views about how to successfully navigate the transition to a world with powerful AI systems. The strategic visions discussed here particularly aim to address three severe risks: **takeover** by powerful misaligned AI systems, **wars** resulting from competitive dynamics around AI, and AI-enabled **concentration of power** among a small group of people.

The goal of this report is to help policy entrepreneurs choose which strategic visions to embrace and identify policy objectives that are robust to uncertainty about which visions are best. Figure 1 shows this logical progression.

- **We first outline nine distinct strategic visions**, discussing their feasibility and how they would address the three covered risks (section B; see Table 1 overleaf). These visions vary along several dimensions, including the type and extent of government involvement, the level of centralization of AI development, and the geopolitical dynamics around powerful AI.
- **We then discuss the desirability of different policy objectives according to various strategic visions** (section C). Some policy objectives, such as improving AI developer cybersecurity, are desirable across almost all strategic visions. However, some objectives are more sensitive to different visions. For example, increasing the U.S.'s lead in advanced AI over China looks less valuable in visions that attempt to preserve an international balance of power.
- **Finally, we discuss the "cruxes" between the different visions** (section D). These are empirical uncertainties that affect how achievable and desirable various strategic visions are, such as the pace of AI progress and how geopolitically transformative AI will be.

Figure 1: The relationship between cruxes, strategic visions, and policy objectives: views on key cruxes help determine strategic vision(s) of choice, which in turn inform policy priorities.

Table 1 – Overall subjective ratings of the nine strategic visions.

Colors represent our assessment of how well each vision addresses each risk and how feasible each vision is to implement: red is high-risk/low-feasibility, green is low-risk/high-feasibility, and yellow is in-between.

Strategic vision	Takeover	War	Power concentr ation	Feasibility
1. Competing private projects: The status quo continues, with many companies competing under light-touch government regulation.				
2. Single private project: One company builds a dominant lead and shapes the trajectory of artificial superintelligence (ASI) unilaterally.				
3. Global private project: A single internationalized company develops AI at the frontier, with financing, talent, compute, and energy infrastructure sourced from many countries.				
4. U.S. leadership with domestic regulation: Private development continues with meaningful government oversight within the U.S., and export controls limit foreign AI competition.				
5. U.S. centralized government project: The U.S. government takes direct control of frontier AI development.				
6. U.S. + allies government project: Similar to the centralized U.S. project, but with some U.S.-allied governments also involved in the project.				
7. International competition and deterrence: Rival powers each run national projects engaged in a managed competition, with mutual deterrence constraining reckless actions.				
8. Global centralized government project: Major powers jointly run a single AI project with shared control and a strong safety focus.				
9. Global coordinated regulator: An international body oversees frontier AI development in all key jurisdictions, ensuring safety.				

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A. Introduction

There are many competing views about how best to manage risks from advanced AI systems. As well as debating each policy question on its merits (export controls, regulatory approaches, etc.), it can be useful to step back and examine the broader *strategic visions* that inform these positions. This report compares nine such visions, i.e., high-level views about how to successfully navigate the transition to a world with powerful AI systems.

Strategic visions often make different assumptions about the nature of AI development, and about the context in which the most advanced AI systems should be developed. For example, the strategic visions that an individual prefers might be affected by that individual's beliefs about the fundamental difficulty of ensuring that advanced AI remains under human control, or about the promise of international cooperation.

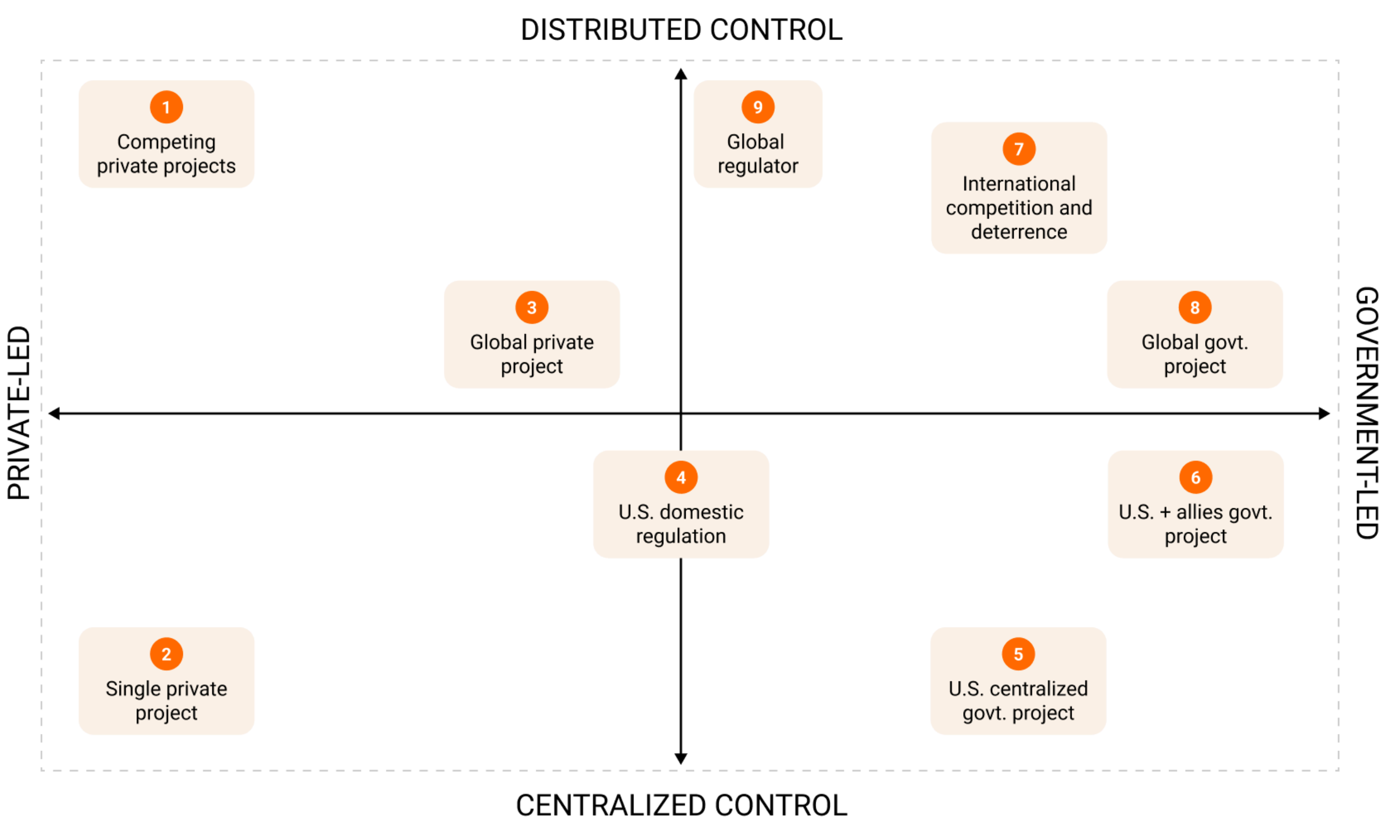

Figure 2: The nine strategic visions mapped to the private/government and distributed/centralized axes.

There are many such assumptions that distinguish which strategic visions an individual might prefer—we provide an overview of some of them in Section D. To organize this landscape, we propose two key axes that capture much of the differentiation between visions:

- **Centralized or distributed control**: At one extreme, artificial superintelligence (ASI) is developed singlehandedly, granting the developer tremendous power over world affairs. Alternatively, power could be distributed among multiple actors or within pluralistic decision-making structures that prevent excessive concentration of power.
- **Private or government control**: Presently, private companies lead frontier AI development. Over time, however, governments may assume increasing control, such as by nationalizing AI developers or forming public-private partnerships.

Each strategic vision stakes out a position on these two axes. Figure 2 places the nine strategic visions we identified in the literature on these axes.

Scope

In this report, we focus on strategic visions for navigating a transition to a world with ASI in a way that reduces several severe risks. ASI refers to AI systems that significantly outperform humans across a wide range of cognitive domains. ASI would pose major governance challenges. For example, keeping such a capable system under meaningful human control might be extremely difficult.[1] Moreover, some models predict flywheel effects, where advanced AI systems could be used to rapidly develop even more advanced AI systems.[2] This means that ASI might be developed at a time of particularly rapid technological change; governance efforts might struggle to keep up.

The severe risks in scope are as follows:

- **AI takeover**: The risk that advanced but misaligned AI systems permanently disempower humanity.[3] This could occur if AI systems develop goals that their operators did not intend and have sufficient capabilities to resist human correction or shutdown.[4] We refer to technical efforts to reduce this risk as "alignment and control efforts."
- **War**: The risk that geopolitical competition or misperception over AI capabilities escalates into great power conflict. For instance, the race to develop advanced AI could trigger preemptive strikes, arms race instability, or crises stemming from misattributed AI actions.[5]

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[1] Bales et al., "Artificial Intelligence: Arguments for Catastrophic Risk."
[2] Davidson et al., "Three Types of Intelligence Explosion."
[3] Dung, "The argument for near-term human disempowerment through AI."
[4] Stix et al., "The Loss of Control Playbook: Degrees, Dynamics, and Preparedness."
[5] Guest and Delaney, "AI, the spectre of decisive advantage, and international conflict."; Mueller, "Heeding the Risks of Geopolitical Instability in a Race to Artificial General Intelligence."

- **Extreme power concentration**: The risk that control over advanced AI systems becomes excessively centralized, undermining pluralism, democratic accountability, and the common good. A single entity—whether a company, government, or small group—could wield ASI to entrench its power indefinitely, foreclosing society's ability to course-correct.[6]

We chose these three risks primarily because they would be particularly severe and irrecoverable if they come to pass.[7] These three risks are, of course, not the only risks associated with advanced AI. Future work could assess how well the various strategic visions handle a broader set of risks. Additionally, it would be valuable for further work to assess the various visions with more of a focus on how well they would also capture upsides associated with advanced AI.[8]

Key concepts

Several key concepts recur throughout this report. **AI timelines** refer to forecasts for when particular milestones, such as artificial general intelligence (AGI), will be developed. **Takeoff dynamics** describes how quickly AI capabilities may improve after AGI is reached—ranging from slow (years or decades to get to ASI) to fast (months). Fast takeoff could result from an **intelligence recursion**,[9] where AI systems that have automated AI R&D can design and build yet-more-capable successors. If only one actor initially achieves this intelligence recursion, they could end up with a very large lead in AI development. The large lead resulting from this recursion might end the **race dynamics** currently in play, where competitive pressures incentivize frontier AI developers to corner-cut on safety and deploy ever more advanced models to win market share and revenue.[10] In particular, this would allow the leading actor to devote more resources to alignment and control efforts, without worrying about a competitor overtaking them.[11] This is

[6] Csernatoni, "Can Democracy Survive the Disruptive Power of AI?"; Hadshar, "Extreme power concentration."

[7] Human misuse of AI systems could also lead to a severe and irrecoverable scenario, such as a terrorist group using AI to develop and release a pandemic-causing pathogen. We do not focus on this class of risks because similar solutions are broadly applicable across most strategic visions: train AI models to refuse dangerous queries [Bai et al., "Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback."], avoid publicly deploying AI systems with the most problematic dual-use capabilities [METR, "Common Elements of Frontier AI Safety Policies."], and use defense-in-depth to harden society against misuse [Snyder-Beattie, "The Four Pillars: A Hypothesis for Countering Catastrophic Biological Risk."]. There are some differences, such as in the relative emphasis on nonproliferation of dangerous technology versus widespread deployment of defensive technology, but misuse defense does not rely as much on the geopolitical structure and level of government involvement in the strategic vision, so we do not focus further on misuse risks here.

[8] As an example of this approach, see Forethought's "Better Futures" series by Will MacAskill.

[9] This is sometimes also referred to as an "intelligence explosion" or "recursive self-improvement."

[10] Emery-Xu et al., "Uncertainty, Information, and Risk in International Technology Races."

[11] Dario Amodei, CEO of Anthropic, writes that "the intensity of the race will make it increasingly hard to focus on addressing autonomy risks". [Amodei, "The Adolescence of Technology"]

sometimes referred to as paying an **alignment tax**, as building robustly aligned models requires additional resources over and above building models aimed solely at being very capable.[12]

Report structure

In section B, we introduce the nine strategic visions and summarize why proponents of each vision think that may be the best bet for navigating the transition to a world with ASI. Specifically, we consider the feasibility of each vision and how they address risks of AI takeover, war, and power concentration. We also consider key criticisms of each strategic vision. Section C discusses important AI policy debates and the contrasting positions different strategic visions take on each. Finally, section D considers the evidential basis for different strategic visions by collating key cruxes that determine which visions are more feasible and desirable.

This report summarizes and responds to a wide body of work. We have endeavored to fairly represent the various views, but the original authors proposing different strategic visions may not always agree with our gloss on their work.

[12] Leike, "Distinguishing three alignment taxes."

B. The landscape of strategic visions

This section maps the major strategic visions currently proposed for navigating the transition to a world with ASI. We informally surveyed published literature to find prominent strategic visions proposed by researchers and policy practitioners.[13] Combining and clarifying overlapping strategic visions left nine main distinct possibilities: three each in the overarching categories of private company-led AI development, U.S. government-controlled development, and internationalized control.

For each vision, we first describe its core logic, then analyze how it addresses the risks of takeover, war, and power concentration, and finally assess its feasibility and desirability. Feasibility refers to whether a vision could realistically be implemented. Desirability refers to whether a vision would lead to good outcomes if implemented.

1. Competing private projects

In this vision, AI would continue to be governed similarly to the status quo, where it is treated as a "normal technology."[14] Many private AI developers (mostly in China and, especially, the U.S.) compete to develop and market the best models, with relatively little government intervention or geopolitical negotiation. This is the main vision advocated for by much of Silicon Valley.[15]

How this vision handles key risks

AI takeover: This strategic vision views keeping AI systems under human control as continuous with regular product safety, where iteration and experimentation with real-world deployment are key. Moreover, this vision relies on alignment being relatively easy to achieve, even for superhumanly capable systems—requiring modest investment rather than fundamental breakthroughs. Under this assumption, market signals from customers create sufficient demand for robustly aligned models, and normal competitive incentives drive companies toward safety rather

[13] See e.g., useful reviews in Maas, "Advanced AI governance: A literature review of problems, options, and proposals" and Maas and Villalobos, "International AI Institutions: A Literature Review of Models, Examples, and Proposals."
[14] Narayanan and Kapoor, "AI as Normal Technology."
[15] E.g. Zuckerberg, "Personal Superintelligence."; Andreessen, "Why AI Will Save the World."; Patel, "Contra Marc Andreessen on AI."; Patel, "Mark Zuckerberg — AI will write most Meta code in 18 months."; Nadella et al., "AI for startups."

than away from it. Advances in coordination and commitment technology could allow companies to make binding agreements to reduce risks without government enforcement.[16]

War: On this vision, AI does not become extremely geopolitically important, e.g., perhaps less important than oil was in the 20th century. So great powers may enter conflicts for normal reasons, and AI may be integrated into militaries, but AI is not seen as a key driver of war.

Power concentration: The AI industry continues to be competitive, and normal antitrust enforcement is sufficient to maintain this market competition.

Feasibility and desirability

Feasibility: This vision matches the status quo, so it might not require active intervention to achieve. However, it assumes no major crisis or government intervention disrupts the current market-led trajectory, and that AI companies will continue to operate with relatively light regulation.

Key advantages:

- A vibrant ecosystem of AI developers and application companies means competition keeps prices down, and consumers reap a large surplus of rapid, beneficial innovation, including in medicine.[17]
- Companies may be more knowledgeable than governments about the nature of risks and how to best mitigate them, and may find better solutions than those arising from prescriptive regulations.
- No novel governance structures are needed, which reduces the room for unforeseen difficulties.

Key disadvantages:

- This vision assumes alignment and control are tractable with modest investment. If alignment and control do require significant investment, competitive dynamics may make coordination difficult. Without government enforcement mechanisms, individual companies face pressure to underinvest in alignment and control relative to capabilities, since the safety benefits are broadly shared while the costs are borne privately.[18]

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[16] Vaintrob et al., "AI Tools for Existential Security."
[17] Of course, these benefits will first flow to rich people and countries. However, medical and other innovations generally benefit lower income people as well, after some lag. And in worlds of radical abundance, (absolute) poverty may be eliminated anyway.
[18] Felstead, "Enabling Frontier Lab Collaboration to Mitigate AI Safety Risks."; Andrews et al., "Collective Action Problems."; Askell et al., "The Role of Cooperation in Responsible AI Development."

2. Single private project

One company automates AI R&D before its competitors and develops an unassailably large lead via an intelligence recursion. Governments are too slow to play a key role in governing the development of ASI. This single company accrues extreme influence and power globally. No major actors publicly advocate for this vision today,[19] but it is useful to include as a logical extreme of unregulated private AI development, to compare other visions against.

How this vision handles key risks

AI takeover: If only one developer will ultimately matter, then avoiding takeover depends primarily on that developer's alignment and control efforts. This plan requires the leading developer to have a sufficient lead over competitors that it can afford to slow its automated AI R&D and pay the alignment tax, investing substantial resources in alignment and control rather than extending its capabilities advantage.[20] Success depends on two factors: the leading developer's alignment and control efforts (including automated efforts) outpacing the difficulty of these problems, and maintaining a lead large enough to absorb the cost.[21]

War: AI does not become geopolitically extremely salient until after one company has an unassailable lead. This company may then be sufficiently diplomatically and strategically skilled to play countries off against each other without a war, while improving its own position and power.

Power concentration: This view explicitly rejects power concentration as something that must be avoided. Instead, on this view, it is imperative that power is concentrated in the right hands, and that the winning company has good values and governance structures that enable a positive future for the world's population despite minimal democratic oversight.

Feasibility and desirability

Feasibility: This scenario currently appears unlikely. Multiple well-resourced companies are near the frontier,[22] whereas this vision would require one to acquire a significant lead. The increasing prominence of advanced AI in public discourse also makes it seem less likely that governments would be so removed from how advanced AI is developed and used.

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[19] That said, some early outputs from the Machine Intelligence Research Institute (MIRI) discussed this plan, e.g., Yudkowsky, "Artificial Intelligence as a Positive and Negative Factor in Global Risk." And some actors within leading AI companies may implicitly aim for this sort of outcome. See, e.g., Hao, "Empire of AI," which argues OpenAI has grown more powerful than almost any nation-state and is engaged in consolidating political as well as economic power.
[20] Aschenbrenner, "Situational Awareness", Section IIId: The Free World Must Prevail.
[21] AI Alignment Forum, "Alignment Tax."
[22] Epoch AI, "AI Benchmarking."

Key advantages:

- This vision sidesteps the difficulty of getting domestic regulation and international negotiations right. Arguably, AI companies are best placed to self-govern, given that governments do not understand the technical nuances well enough.
- Competitive pressures to deploy models quickly are minimized by one company having a large lead.

Key disadvantages:

- Concentrating control over advanced AI in a single private entity raises serious legitimacy concerns. A small group of individuals, selected by market dynamics rather than any deliberate process, would be making civilization-shaping decisions without democratic mandate or broad accountability.[23]
- Even setting aside legitimacy, concentrating decision-making authority in a small group may produce worse outcomes for humanity.
- Developers pursuing a decisive lead over competitors could intensify racing dynamics across the industry. If other developers perceive that falling behind means losing any influence over advanced AI, they may cut corners to keep pace.[24]

3. Global private project

A single multinational company does frontier AI development, with financing, talent, compute, and energy infrastructure sourced globally. This could happen if long AI timelines and increasing upfront costs of pretraining mean that there is natural consolidation of the AI industry into one leading company. Governments would likely be somewhat involved, such as through investments from sovereign wealth funds, and granting permissions for mergers and acquisitions despite antitrust issues. Optionally, governments may ban rival private AI development to prevent the internationalized project from needing to race. But the project is a private company with standard corporate governance (such as shareholders and a board of directors). This vision was proposed by Nick Bostrom.[25]

How this vision handles key risks

AI takeover: Similar to the single private project vision, one project develops a large lead (in this case due to market consolidation and possibly government intervention to create or condone an internationalized AGI project). This project uses its lead to devote significant resources to alignment and control efforts, without worrying about a competitor overtaking it.

[23] LaForge et al., "Power and Governance in the Age of AI."
[24] Emery-Xu et al., "Uncertainty, Information, and Risk in International Technology Races."
[25] Bostrom, "Open Global Investment as a Governance Model for AGI."

War: In this vision, major powers are invested (financially and otherwise) in this project's success, and so will not be incentivized to sabotage the project or intervene militarily. Economic and technological interdependency could also reduce the risk of major wars.

Power concentration: The single internationalized project would accrue immense power in this model. However, due to the broader set of countries and institutions involved in the governance of the project, it is significantly less likely that a few company leaders will centralize immense power personally.

Feasibility and desirability

Feasibility: To the extent that this vision involves international cooperation, it may be more challenging than purely domestic visions, since cooperation between adversaries on strategic technologies is challenging. However, Bostrom explicitly proposes this vision as a more feasible alternative to other plans for AI governance at the international level that would require formal treaties and more novel kinds of international institutions.

Key advantages:

- Multinational corporations are a well-known legal structure with tried and tested governance mechanisms, though it is unclear whether these mechanisms will be robust to the extreme power the company may come to have.
- The internationalized project will have the resources and motivation to invest significantly in risk management, since this project will internalize a large share of the benefits and is not in an intensive competition.

Key disadvantages:

- Despite multiple actors being involved in the project's governance (e.g., via board seats), the company constitutes a single point of failure.
- Investors may be shortsighted and prioritize rapid monetization of the project's technology over taking careful efforts to reduce risks.

4. U.S. leadership with domestic regulation

Multiple private frontier AI developers continue operating in the U.S., but the government becomes alert to the strategic significance and security risks from AI, so it plays a more active role in ensuring safe AI development. Internationally, the U.S. leverages export controls and diplomacy to constrain competing foreign projects and maintain strategic AI dominance. This vision preserves the innovative capacity of the private sector while introducing credible regulation to mitigate catastrophic risks. Anthropic's Dario Amodei, among others, has expressed support for versions of this vision, emphasizing both safety and the need for U.S. dominance to preserve democratic values.[26]

How this vision handles key risks

AI takeover: According to this vision, the U.S. government can wisely regulate AI development to enforce safety standards that are sufficient to prevent AI takeover, without being so onerous that they significantly degrade the U.S.'s AI advantage. For instance, the government may require companies to produce a compelling safety case[27] before training a frontier model, and undergo thorough third-party evaluations before publicly deploying a model.

War: AI development is still led by competing private companies in this vision, so foreign adversaries might be less worried than in cases where the U.S. government nationalizes or militarizes AI development in a centralized project.[28] In this vision, international AI competition is primarily economic rather than military. However, U.S. government efforts to secure AI supremacy, including strict export controls, could be viewed by other countries as challenging their core interests.[29] A sufficiently extreme version of this concern might motivate military action to protect these interests.

Power concentration: This view explicitly aims to concentrate power in the U.S. However, domestically, power is distributed between multiple AI developers, each in turn checked by the government. This vision relies on companies and the government balancing each other's power, maintaining the benefits of both free-market competition and democratic oversight.

Feasibility and desirability

Feasibility: This vision would require significant political will in the U.S. to meaningfully regulate AI to reduce risks. Such will is not currently in place.[30] However, it would require much less significant political changes than some of the other visions, and the U.S. government is already committed to part of this vision, such as with efforts to slow AI development by U.S. adversaries.

Key advantages:

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[26] Amodei and Pottinger, "Trump Can Keep America's AI Advantage."; Toner, "Senate Judiciary written testimony - Helen Toner.", p. 5; The Chips for Peace proposal is similar, with more emphasis on U.S. cooperation with allies [O'Keefe, "Chips for Peace: How the U.S. and Its Allies Can Lead on Safe and Beneficial AI"].

[27] A structured argument, supported by evidence, that the system will operate safely under specified conditions; Clymer et al., "Safety Cases: How to Justify the Safety of Advanced AI Systems."

[28] Anderson-Samways, "The US Government's Role in Advanced AI Development: Predictions and Scenarios."

[29] Shivakumar et al., "Balancing the Ledger: Export Controls on U.S. Chip Technology to China."

[30] Vance, "Remarks by the Vice President at the Artificial Intelligence Action Summit in Paris, France."

- Multiple competing private developers prevent any single company from becoming a concentrated point of failure, while government regulation provides an additional check on corporate power. This balances private innovation with democratic oversight.
- Keeping AI development primarily in the private sector reduces the military framing that could make U.S. AI dominance appear more threatening to adversaries, potentially reducing escalation risks compared to centralized government projects.

Key disadvantages:

- The federal government is reluctant to regulate AI developers, so without a crisis or other major political change, government oversight and minimum safety standards may be very weak.
- Continued racing between private AI developers could incentivize merely perfunctory compliance with requirements, allowing significant AI takeover risks to persist.

5. U.S. centralized government project

The U.S. government decides that AI is so strategically important that its development should not be left to private companies, and it therefore centralizes U.S. AI development under government control. This could take the form of a government-led consortium with public-private partnerships, modelled on past efforts such as the Manhattan Project or the Apollo Program.[31] This U.S. government-led project develops a large lead over international competitors and independent domestic AI projects (if any),[32] and uses this lead to deploy AI systems safely and carefully without needing to race to capture market share. This view was proposed in *Situational Awareness* and also by the U.S.-China Economic and Security Review Commission.[33]

How this vision handles key risks

AI takeover: Centralizing U.S. AI development means that the project need not race against other AI developers domestically, allowing for greater investment in alignment and control efforts.

War: U.S. government centralization of AI development may be viewed by adversaries as threatening. This would especially be the case if the Department of War housed or was majorly involved in the newly centralized AI project.[34] This vision relies on U.S. strength deterring

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[31] Anderson-Samways, "The US Government's Role in Advanced AI Development: Predictions and Scenarios."; Chessen and Martell, "Beyond a Manhattan Project for Artificial General Intelligence."
[32] For instance, the government could nationalize or ban competing projects as a last resort if they are not willing to join a government-led consortium.
[33] Aschenbrenner, "Situational Awareness," Section IV: The Project.; U.S.-China Economic and Security Review Commission, "2024 Annual Report to Congress."
[34] Anderson-Samways, "The US Government's Role in Advanced AI Development: Predictions and Scenarios."

adversaries and imposing a Pax Americana. This could include supplying cooperating countries with subsidized access to advanced U.S. AI systems or sharing other benefits of AI development (such as resulting medical innovations) with the world.[35]

Power concentration: This vision embraces centralization of power within the U.S. government, and relies on Constitutional checks and balances and democratic oversight to maintain good societal outcomes.

Feasibility and desirability

Feasibility: This vision requires a major expansion of federal authority over a private industry and the political willingness to nationalize or tightly control leading AI companies. It is more likely to occur in response to a major AI incident or national security crisis, or if AI development costs become too large for private actors.

Key advantages:

- This vision foregrounds democratic oversight of AI development (through the government) more than visions focused on private, and relatively unaccountable, AI developers.
- A centralized project having a large lead allows alignment and control efforts to take precedence over rushing new models to market.

Key disadvantages:

- There is a notable risk that China or other adversaries perceive such a project as aggressive and respond with an accelerated centralized project of their own, or with efforts to disrupt the U.S. project. This could mean that the U.S. centralized project exacerbates race dynamics.[36]
- The radical increase in the power of the federal executive branch from advanced AI may threaten constitutional democracy by making it harder for other parts of government to impose checks and balances.

6. U.S. + allies government project

The U.S. leads a coalition of allies and partners in an intergovernmental AI project, in an internationalized version of the previous vision.[37] Relevant allies and partners could include EU member states, the UK, Canada, Australia, Japan, South Korea, and Taiwan. Collectively, these countries have a very dominant position in semiconductor manufacturing,

[35] Clare et al., "Options and Motivations for International AI Benefit Sharing."
[36] Katzke and Futerman, "The Manhattan Trap: Why a Race to Artificial Superintelligence is Self-Defeating."
[37] A U.S. + allies project could arise after an initial U.S.-only project is set up, or allies could be included from the outset.

research talent, and financing capacity, making it easier to outpace competing national efforts than if the U.S. acted alone.[38] These allies could pool resources to jointly develop AI safely and securely, with benefits from and control over resulting AI systems shared between member countries (perhaps in proportion to their financial and in-kind contributions to the project).[39] Such a project could draw inspiration from Intelsat,[40] or even the original Manhattan Project's inclusion of Canada and the UK.[41]

Proactively including allies and partners in a project would likely reduce the chance that these countries would launch their own sovereign AI projects. This would have various effects on the risks discussed in this report, though it is unclear what the overall direction would be. For example, consolidation might reduce takeover risk by decreasing the number of independent AI projects, each of which represents a potential point of failure. On the other hand, it might increase the risk of concentration of power, since it could reduce the number of advanced AI projects.

How this vision handles key risks

AI takeover: As with the U.S.-only centralized project, having a large lead allows for greater alignment and control efforts without worrying about competitors catching up. The U.S. + allies project could have an even more dominant lead, allowing even greater efforts, if needed.

War: On this vision, the world order would remain U.S.-led, but with greater emphasis on cooperation among allies rather than unilateral U.S. action. The involvement of multiple countries in AI development could have a stabilizing effect. Joint decision-making among allies tends to be more deliberate and constrained than unilateral action, which may reassure rival nations that the coalition would not take sudden, aggressive action against them, reducing the perceived need for preemptive action.[42]

Power concentration: AI development would still be concentrated in a single institution, but this institution would be controlled by a broader set of countries.

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[38] Belfield, "Domestic frontier AI regulation, an IAEA for AI, an NPT for AI, and a US-led Allied Public-Private Partnership for AI: Four institutions for governing and developing frontier AI," pp. 28-30.
[39] A U.S.+ allies project may be especially likely if AI timelines are long and the financial and energy costs of frontier training become prohibitive for the U.S. to bear alone.
[40] MacAskill and Hadshar, "Intelsat as a Model for International AGI Governance."
[41] Wellerstein, "Conant on the Role of the British in the Manhattan Project."
[42] Belfield, "Domestic frontier AI regulation, an IAEA for AI, an NPT for AI, and a US-led Allied Public-Private Partnership for AI: Four institutions for governing and developing frontier AI," p. 31.

Feasibility and desirability

Feasibility: This vision requires substantial coordination among the participating countries. This might be challenging if the U.S. were reluctant to include other countries in its hypothetical centralized AI development project, especially if allies do not have much bargaining power.

Key advantages:

- An intergovernmental project increases the chance that overall democratic oversight of the project is maintained, since no one country is in sole control.
- A joint project can build up a large lead over all rivals, making it easier to devote significant resources to risk reduction.

Key disadvantages:

- As with the U.S. centralized project, there is still some (though mitigated) risk from centralization of power within one project.

7. International competition and deterrence

A stable equilibrium emerges where two or more geopolitical blocs pursue AI projects, but do not race towards ASI for fear of provoking adversaries into military strikes.[43] Each side may be so worried about the other achieving ASI and a decisive strategic advantage that they explicitly or implicitly threaten to sabotage projects recklessly racing to ASI.[44] The competing government projects could coordinate to slow down AI progress until ASI can be developed safely.[45] The future is then shared between the rival powers, such as the U.S. (+ allies) and China. This mutual assured AI malfunction (MAIM) deterrence equilibrium was proposed by Hendrycks, Schmidt, and Wang in *Superintelligence Strategy*.[46]

How this vision handles key risks

AI takeover: On this vision, mutual deterrence prevents all-out racing towards ASI, allowing more time for AI developers to work on alignment and control techniques.

War: AI is central to geopolitical competition in this vision, but war can be avoided by all sides rationally calculating that managed competition is preferable to escalation.

[43] Delaney, “Crucial Considerations in ASI Deterrence.”
[44] Chase and Marcellino, “Incentives for U.S.-China Conflict, Competition, and Cooperation Across Artificial General Intelligence’s Five Hard National Security Problems.”
[45] Abraham et al., “A Prisoner’s Dilemma in the Race to Artificial General Intelligence.”
[46] Hendrycks et al., “Superintelligence Strategy.”

Power concentration: This vision maintains a multipolar world order during and after the development of superintelligence, thereby avoiding complete centralization of power. However, it is still possible that within the competing blocs, power will be greatly centralized.

Feasibility and desirability

Feasibility: A MAIM-style deterrence equilibrium would depend upon three premises being correct: that adversaries would expect to significantly lose out from adversaries' AI development, that this would motivate them to make threats, and that countries would be influenced by these threats. One of us (Delaney) discussed these premises in detail, concluding that each was somewhat more likely than not.[47]

Key advantages:

- Countries have stronger incentives to reduce takeover risk because this risk could motivate other countries to take aggressive action against them.
- This vision might balance power between multiple countries.

Key disadvantages:

- Those who prioritize having liberal democratic values shape the post-ASI world may find this vision concerning, as it preserves significant influence for states that are non-democratic but have significant bargaining power via their AI or military capabilities.
- Although intended to avoid war, deterrence equilibria could be unstable, such as if different countries have different understandings of what actions would be unacceptable.

8. Global centralized government project

Major powers agree to consolidate frontier AI development into a single internationally governed project.[48] The U.S., China, and possibly other countries would jointly control this effort. This would mean that states would be jointly developing advanced AI, instead of competing with each other to do so. Proposals along these lines include some CERN for AI proposals,[49] the Multilateral AGI Consortium proposal,[50] and proposals modeled on the Baruch Plan[51] (which analogously aimed to reduce risks from nuclear weapons by placing them under international control).[52]

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[47] Delaney, "Crucial Considerations in ASI Deterrence."
[48] Another way this vision could arise is via first having a global private project and then increasing intergovernmental oversight and control over that initially private project.
[49] Kohler, "CERN for AI: One Analogy, Many Visions."; Petropoulos et al., "Building CERN for AI."
[50] Hausenloy et al., "Multinational AGI Consortium (MAGIC): A Proposal for International Coordination on AI."
[51] CBPAI, "Why a Baruch Plan for AI?"; Zaidi and Dafoe, "International Control of Powerful Technology: Lessons from the Baruch Plan for Nuclear Weapons."
[52] See also Hogarth, "We must slow down the race to God-like AI" and Miotti et al., "A Narrow Path."

How this vision handles key risks

AI takeover: As with other centralized visions, a single project can build a large lead and devote substantial resources to alignment and control efforts without needing to worry about competitors overtaking it.

War: All major AI/military powers are included in this project, and so provided the project does not fragment, there is low risk of war caused by major powers fearing their rivals having much more advanced AI systems than them.

Power concentration: This arrangement diffuses power more broadly internationally, but this power is still channeled through the centralized AI project, creating a single point of failure. Robust governance structures would be important to prevent regulatory capture or a de facto slide toward world government.

Feasibility and desirability

Feasibility: This is among the most ambitious visions, requiring close cooperation between rival countries on strategically critical technology. However, proponents might believe that it could emerge from growing mutual recognition that uncoordinated development poses unacceptable risks, or from a prolonged MAIM-style standoff where formalization becomes attractive to both sides. This scenario is especially unlikely if ASI arrives soon, since it is far outside what is politically achievable today.

Key advantages:

- A global AI project could achieve buy-in from all major powers and therefore minimize the chance of AI-related wars. By circumventing race dynamics, the centralized project would have more opportunity to invest in alignment and control without being outcompeted.

Key disadvantages:

- Including autocracies in this international project would preserve their influence over global affairs, which may be undesirable compared to a potential alternative future where liberal democracies collectively control ASI and therefore shape the world order.

9. Global coordinated regulator

An international regulator has insight into and oversight of all frontier AI projects. Multiple private developers or governments could still pursue frontier AI development, but would be blocked from further accelerating AI capabilities research if the international regulator deems that they have not demonstrated the adequacy of their alignment and control efforts. The international regulator must have some verification and enforcement power to ensure it is not simply ignored. Such proposals, sometimes called an “IAEA for AI,” have been made by OpenAI, among others.[53] MIRI has proposed a more radical version of a global regulator that would enforce a ban on AI development above some capabilities or compute threshold, until enough progress has been made on reducing the risk of takeover.[54]

How this vision handles key risks

AI takeover: In the best case of an extremely technically competent global regulator, risks from AI development could be accurately assessed and unduly dangerous development prevented, significantly reducing takeover risk. If there are many competing AI projects, the regulator also needs to be sufficiently confident that no project will fail catastrophically, since even one failure may lock in a bad overall outcome. Crucially, the regulator must also have credible mechanisms to enforce its decisions. This could include inspection powers, the ability to impose sanctions, or even authority to physically intervene in non-compliant projects, though establishing and maintaining such enforcement powers at the international level poses significant political challenges.

War: If countries trust the regulator to prevent AI development that poses significant accident risks, then this vision could reduce the likelihood of war, since countries would not have reason to escalate due to fears of an accidental AI catastrophe. However, merely ensuring that AI development is “safe” (in the sense of not posing accident risks) might not fully reassure countries. They might still worry that “safe” AI systems would be *deliberately* used against them by rivals.[55]

Power concentration: This strategic vision does not require frontier AI development to be concentrated into a single entity, so it poses a lower risk of excessively concentrating power.

Feasibility and desirability

Feasibility: This scenario would be even more politically difficult than the (already challenging) prospect of comprehensive U.S. domestic AI regulation. Significant challenges remain, including how best to verify international AI capabilities, convince major powers to allow visibility into their AI projects, and enforce rulings on non-compliant states. However, this vision requires less trust than full consolidation, since AI development remains distributed.

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[53] Altman et al., “Governance of superintelligence.”; Emery-Xu et al., “International governance of advancing artificial intelligence.”; Wasil, “Do We Want an “IAEA for AI”?”
[54] Scher et al., “An International Agreement to Prevent the Premature Creation of Artificial Superintelligence.”
[55] A potential solution would be for the regulator to ensure that AI development is not just safe but also happens in a way that produces AI systems that could not be used for aggressive actions on the international stage. In “Demonstrating restraint,” Patell and Guest provide an overview of mechanisms that could be used here but note that they are often speculative or would require technologies that are yet to be developed.

Key advantages:

- This vision shares the benefits of a U.S. regulator governing private companies, while also providing more international buy-in and legitimacy.

Key disadvantages:

- An international regulator is subject to political capture and could be used for purposes other than to reduce the risks described above.[56] For instance, countries may pressure the regulator to quickly approve AI models from that country, while delaying or denying approval for rival countries.

Transitioning between strategic visions

Figure 3 summarizes some of the most plausible transition pathways between strategic visions, as described throughout this section.

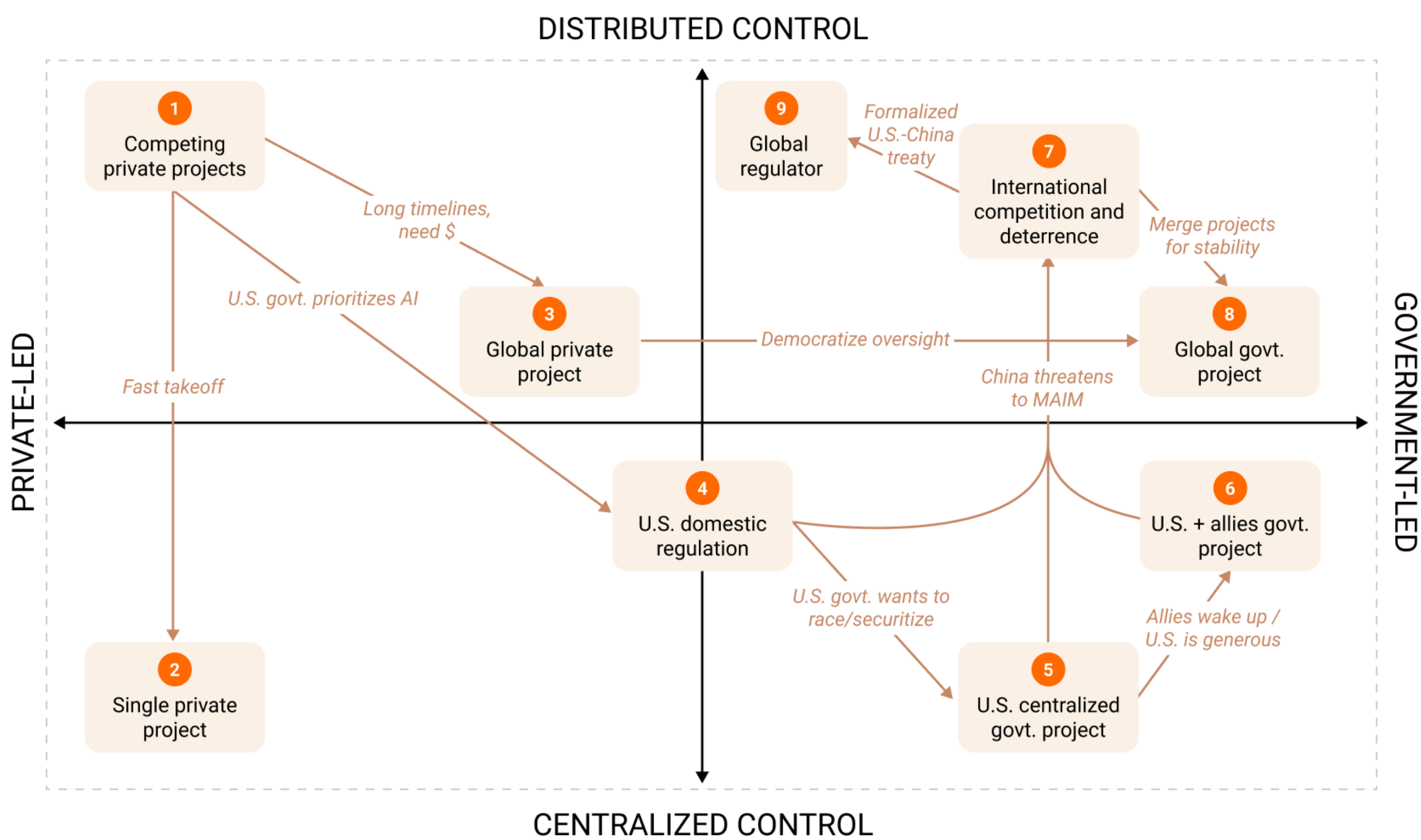

Figure 3: Plausible pathways and causes by which we may transition between the nine strategic visions.

[56] Guest, "Undue influence in an international AI regulator."

C. Policy implications of strategic visions

If all these competing strategic visions differ at the conceptual level but end up recommending similar policies regardless, there would be little point in considering this taxonomy. In practice, however, different visions often propose quite diverging policies. In this section, we assess six illustrative examples of plausible policy goals where there is divergence among competing strategic visions about whether the goal is desirable. For each policy goal, we give an example policy to further this goal and suggest which strategic visions are in favor of or against pursuing this goal (summarized in Table 2).

Strategic vision	Cyber-security	Trans-parency	U.S. govt. focus on AGI	U.S.-China coordination	U.S. lead over China	Pacing AI development
1. Competing private projects						
2. Single private project						
3. Global private project						
4. U.S. domestic regulation						
5. U.S. centralized govt. project						
6. U.S. + allies govt. project						
7. International competition						
8. Global govt. project						
9. Global regulator						

Table 2: Policy goals are shown as being supported (green), opposed (red), or neither (yellow) by each strategic vision.

Increase AI developer cybersecurity

Cybersecurity for AI developers encompasses protecting model weights, training data, and algorithmic innovations from theft or sabotage. Key concerns include preventing adversaries from stealing models (confidentiality), preventing tampering with AI systems (integrity), and maintaining reliable access to AI capabilities (availability).[57] In RAND's seminal framework, security levels for AI developers range from basic protections against amateur opportunistic attackers (level 1) up to defenses against sophisticated state-sponsored operations (levels 4 and 5).[58]

Example policy: Require that frontier AI developers meet strict cybersecurity standards (e.g., security level 4/5) to be eligible for U.S. government contracts.

In favor: Most views are in favor of improved cybersecurity. In particular, visions that rely on some actors having a large lead over everyone else—such as U.S. private developers over the rest of the world, or a U.S. (and allies) centralized project—benefit greatly from AI models being difficult to steal.[59]

Against: All views are in favor of security level 3, sufficient to prevent terrorists and rogue actors from stealing models. However, security level 5 could paradoxically increase the risk of physical conflict in internationalized visions. If a rival cannot use cyberattacks to level the playing field or gather intelligence, they may feel cornered into considering more drastic options, such as kinetic strikes on an adversary's data centers. This counterintuitive dynamic is particularly relevant for the MAIM framework, which depends on all parties retaining some ability to disrupt reckless AI development.[60] Strategic visions that could come about because of MAIM dynamics—a global regulator, and a globalized project—could also benefit from not reaching security level 5.

Increase AI developer transparency

Transparency for AI developers can take many forms, including disclosing training data sources and model capabilities, reporting on internal deployment of AI systems, sharing safety testing results, or providing access to external auditors.[61] Greater transparency can enable better

[57] Nevo et al., "Securing AI Model Weights."
[58] Nevo et al., "Securing AI Model Weights."
[59] However, if advanced cybersecurity proves very costly in terms of time or money, the benefits may not be sufficient to warrant investing in level 4/5 security.
[60] Hendrycks et al., "Superintelligence Strategy."
[61] Anthropic, "The need for transparency in Frontier AI."

regulatory oversight and public accountability, though it may also create competitive disadvantages or security risks if sensitive information is disclosed too broadly.[62]

Example policy: Require AI developers to disclose how they are using internal-only models and what risks these models pose.[63]

In favor: Increased transparency is particularly valuable for the U.S. domestic regulation vision, which relies on the government having adequate insight into private AI developers to design and enforce sensible guardrails. Moreover, additional transparency in the status quo of competing private projects could help make the U.S. government more aware of AI risks and thus bring about visions that involve stricter regulations if and when necessary. For internationalized AI projects (private or government), transparency would likely occur by default and would help with shareholder or democratic oversight of the project.

Against: If one favors minimal government involvement and thinks that governments would not be able to interpret companies' AI transparency disclosures well, then additional transparency requirements may actually be bad and slow down private AI developers without reducing risks. However, some amount of voluntary, light-touch transparency could forestall more heavy-handed regulation, and so be good for companies continuing AI development relatively unregulated. Additionally, in a U.S. (+ allies) project, some amount of secrecy may be helpful to reduce the amount of information about the project that is available to adversary countries.

Sharpen U.S. government focus on AGI/ASI

Many policymakers and government officials remain unfamiliar with the technical trajectory and strategic implications of advanced AI. Improving government understanding of and focus on AGI/ASI risks and opportunities could enable better-informed policy decisions, from targeted regulations to potential government involvement in AI development.[64]

Example policy: Briefing Congressional offices and Executive branch staffers on key AGI/ASI topics, such as forecasts for when different capability levels will be reached.

In favor: This policy goal is favored by strategic visions centered on government leadership in AI development. Domestic regulation also benefits from increased government knowledge of and focus on AI. Better-informed policymakers are more likely to allocate appropriate resources towards careful AI risk management and to be prepared to respond effectively in an AI crisis.

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[62] For example, see the discussion in Guest, "Topics for Track IIs: What Can Be Discussed in Dialogues About Advanced AI Risks Without Leaking Sensitive Information?" for risks from sharing certain kinds of information.
[63] Apollo Research, "Internal Deployment of AI Models and Systems in the EU AI Act."
[64] Marsh, "Early US policy priorities for AGI."

Internationalized strategic visions also generally benefit from increased U.S. government focus on AI risks and opportunities, since such international approaches will require U.S. government buy-in.

Against: If one thinks that the government is too technically unsophisticated to competently regulate (or develop in-house) a technology as fast-moving as AI, increasing government focus on AI may backfire. Specifically, there could be an unhappy valley where the government has learned enough to think it can regulate wisely, but in reality, its knowledge is still very limited. Of course, the government is not a monolith, and staff spread across agencies and offices will have very different levels of understanding of AI. So the important question is how informed the writers and implementers of governmental policies around AGI and ASI would be.

Improve U.S.-China coordination on AI risk

U.S.-China coordination on AI could range from informal dialogues to create mutual understanding, to formal agreements on safety standards, to joint research initiatives. Even limited coordination, such as establishing common terminology or sharing information about AI incidents, could reduce misperception and lay the groundwork for deeper cooperation if both sides later see it as beneficial.[65]

Example policy: Engage in a series of Track 1 dialogues to create common knowledge of AI risks and mitigations, and propose areas of cooperation for mutual benefit.[66]

In favor: Strategic visions aiming for international cooperation and standardization view this policy goal as crucially important, as dialogues can form the basis of more extensive cooperation later.

Against: Greater U.S.-China cooperation could make an internationalized approach to ASI governance more likely (e.g., a global regulator or joint project would be more feasible), which could be undesirable if one favors a U.S. (+ allies) dominated future.

Increase the U.S. lead over China in AI

The U.S. currently leads China in frontier AI capabilities by an estimated seven months (though estimates of the gap vary).[67] Policy tools available to U.S. policymakers to maintain or extend this lead include export controls on advanced AI chips and semiconductor manufacturing equipment, immigration policies to attract AI talent, and investments in domestic AI research and compute infrastructure.

[65] Hass and Kahl, "Laying the groundwork for US-China AI dialogue."; Siddiqui et al., "Promising Topics for U.S.-China Dialogues on AI Risks and Governance."
[66] Delaney and Guest, "Practical Provisions for U.S.-China Cooperation on AI Risks."; Siddiqui et al., "Promising Topics for U.S.-China Dialogues on AI Risks and Governance."
[67] Emberson, "Chinese AI models have lagged the US frontier by 7 months on average since 2023."

Example policy: Strengthen export controls on advanced AI chips and semiconductor manufacturing equipment going to China.[68]

In favor: Strategic visions focusing on U.S. AI dominance (either from companies or the government) see creating a larger lead over China as a key goal. This reduces the intensity of the competition to develop advanced AI, allowing the U.S. to invest more in alignment and control efforts without risking a Chinese project catching up.

Against: If one favors more international strategic visions, a large U.S. lead may be counterproductive, as it could encourage the U.S. to pursue unilateral ASI, with associated risks of misalignment, war, and power concentration.

Limiting the pace of frontier AI capabilities advancement

Various policies could deliberately or incidentally slow the rate of frontier AI development. Some have argued that such slowing might be beneficial because this would make it easier for risk-mitigation efforts to keep pace with AI systems becoming more capable.

Example policy: International agreements to limit the pace at which training compute can be scaled up, thereby slowing the rate of frontier capabilities progress.[69]

In favor: Slowing progress towards particularly capable AI systems being developed might be necessary if one wanted to buy time to implement a particularly time-consuming vision. For example, visions that involve significant international cooperation might be challenging without this measure because negotiating international agreements can be slow.

Against: The nearer one's preferred strategic vision is to the status quo, i.e., U.S. corporate-led AI development, the more valuable it might be for AI development to progress quickly before the strategic picture is upturned by technical or geopolitical developments. There are also practical concerns: unilateral pauses may be difficult to verify and could simply shift AI leadership to adversaries or less safety-conscious actors.[70]

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[68] Khan et al., "Should the US Sell Hopper Chips to China?"

[69] See, for example, Miotti, "An international treaty to implement a global compute cap for advanced artificial intelligence."

[70] For example, Vice President J.D. Vance expressed these concerns in an interview: "part of this arms race component is if we take a pause, does the People's Republic of China not take a pause? And then we find ourselves all enslaved to P.R.C.-mediated A.I.?" [Douthat, "JD Vance on His Faith and Trump's Most Controversial Policies."]

D. Cruxes between strategic visions

The future is uncertain. We don't know who will develop key AI capabilities, or when, or how the political and geopolitical landscape will have changed by then. But these and other key questions help determine which strategic visions are most feasible and desirable. In this section, we consider eight such cruxes for choosing a strategic vision where informed AI researchers and policy analysts routinely disagree about what is most likely. For each crux, we provide a very brief survey of notable reasons to believe or disbelieve the claim, and how this would impact one's choice of strategic vision. These are summarized in Table 3. These cruxes are thorny research questions in their own right: we simply aim to touch on some of the most important considerations for and against.

Strategic vision	AGI soon	Fast takeoff	Large lead possible	DSA from ASI	U.S. govt. sees AI as top priority	Intl. coop. feasible	Govts. more trustworthy	Alignment and control very difficult
1. Competing private projects	green	green	yellow	red	red	yellow	red	red
2. Single private project	green	green	green	green	red	red	red	yellow
3. Global private project	red	red	green	green	yellow	green	yellow	green
4. U.S. domestic regulation	green	red	yellow	green	green	yellow	green	yellow
5. U.S. centralized govt. project	red	yellow	green	green	green	yellow	green	green
6. U.S. + allies govt. project	red	yellow	green	green	green	green	green	green
7. International competition	yellow	red	red	yellow	green	green	green	green
8. Global govt. project	red	yellow	green	green	yellow	green	green	green
9 Global regulator	red	red	yellow	red	yellow	green	green	yellow

Table 3: Strategic visions are shown as being supported (green), opposed (red), or roughly neutral (yellow) by these eight possible views on key cruxes, considering both desirability and feasibility. For instance, AGI coming soon makes the "Competing private projects" vision more likely, so that cell is green.

Note that many of these cruxes are correlated. For instance, on the first four cruxes, covering timelines and takeoff dynamics, many people cluster into thinking that short timelines and fast takeoff result in a large lead that confers a decisive strategic advantage, or the opposite, that longer timelines and slower takeoff leave the field more competitive and multipolar. But it is still useful to separate out these nuances, as it is possible to believe some other combination of these, or place significantly different credences in each claim. The short-timelines cluster of views will usually also expect less and worse U.S. government involvement, weaker international coordination, and a higher chance of misalignment.

AGI will come soon

This crux concerns when AI systems will reach human-level capabilities across a broad range of cognitive tasks. We use "soon" here in a general sense to mean roughly within a decade.

Reasons for:

- AI companies, and some (but not all) key AI experts, are predicting AGI will arrive in the next 2–10 years.[71]
- Extrapolating rates of progress from the last several years suggest nearterm AGI is plausible.[72]

Reasons against:

- LLMs in the current paradigm excel at tasks involving learned knowledge and pattern recognition, but still struggle with making novel, important insights and learning on the job.[73]

Implications for strategic visions:

- The sooner one expects AGI to come, the less likely it is that major changes will occur in domestic AI governance and the broader geopolitical environment. In particular, "competing private projects" and "domestic regulation" become more likely on short timelines, while a government centralized project, and especially an international project or regulator, becomes less likely.

ASI will follow shortly after AGI

This crux concerns the gap between human-level AI and superintelligence. A short gap (months to a few years) would leave less time for societal adaptation to any challenges posed by advanced AI.

Reasons for:

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[71] Todd, "Shrinking AGI timelines: a review of expert forecasts."; Toner, "'Long' timelines to advanced AI have gotten crazy short."
[72] Kwa et al., "Measuring AI Ability to Complete Long Tasks."
[73] Patel, "Why I don't think AGI is right around the corner."

- AGIs automating AI research and engineering will greatly increase the effective workforce of AI developers, potentially leading to an intelligence recursion.[74]
- The amount of compute needed to train an AGI-level system will likely be very large. If this AGI designs a novel paradigm that is far more compute-efficient, the same stock of compute could be used to train a far more powerful model.[75]
- Developers of AGI will likely receive enormous revenues, allowing them to invest heavily in continuing their AI research.

Reasons against:

- The first AGI-level systems may be very expensive to run, and so not be that transformative at first.[76]
- Takeoff speed models have many uncertain parameters; automated AI R&D may quickly plateau, as new ideas and innovations get harder to find.[77]

Implications for strategic visions:

- AGI may act as an important catalyst for capturing governments' attention. But governments tend to act slowly, so if ASI follows very quickly after AGI, governments may not have had time to decide on and implement their preferred regulations (or centralization/nationalization of AI development). So rapid takeoff makes it more likely that AI development will remain private-led.
- A longer lag between AGI and ASI also provides more time to use human-level but not-takeover-capable AI systems to make progress on alignment and control,[78] and help with societal coordination and good decision-making.[79]

One actor can build a large lead in AI development

This crux concerns whether AI development will remain competitive among multiple actors, or whether one actor (such as a company or country) can achieve a commanding and durable advantage over all others.

Reasons for:

- There may be a natural monopoly in AI development, where the leading developer achieves a larger market share, collects more customer data and revenue, attracts top talent, and

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[74] Eth and Davidson, "Will AI R&D Automation Cause a Software Intelligence Explosion?"
[75] Pilz et al., "Increased Compute Efficiency and the Diffusion of AI Capabilities."
[76] Ord, "Inference Scaling Reshapes AI Governance."
[77] Erdil et al., "GATE: An Integrated Assessment Model for AI Automation."
[78] Carlsmith, "AI for AI safety."
[79] Finnveden, "What's important in "AI for epistemics"?"

leverages these advantages to build still better models and increase their lead.[80] Automated AI R&D could accelerate this flywheel and build a large lead for the frontier developer.

- The U.S. has a significant advantage in compute, and along with its allies, it controls the semiconductor supply chain, meaning that its compute advantage is likely to persist. This compute advantage could help the U.S. to build a large lead.[81]
- Most of the leading algorithmic innovations have come from the U.S. and allies rather than China.[82] A continuation of this trend would benefit the U.S. lead, if these innovations do not diffuse to other countries.

Reasons against:

- The loosening of export controls, as well as chip smuggling, could allow China to accumulate a significant amount of compute, making for closer competition between the U.S. and China.[83]
- If the U.S. alienates allied countries and makes it harder for skilled workers to immigrate to the U.S., the U.S.'s talent advantage may shrink or reverse.[84]
- Cybersecurity and information security are challenging.[85] This means that sophisticated actors might be able to steal model weights and algorithmic insights, making it hard for any one actor to hold onto a large lead. However, if inference-time compute continues to be a key driver of capabilities, then even two actors with access to the same models may have very different capabilities depending on how many copies they can run.[86]

Implications for strategic visions:

- If the U.S. lead stays strong, this favors U.S. private or government-led development, and makes internationalized approaches less likely.
- If no actor can maintain a large lead, this makes visions with stronger international aspects more necessary, as diffusion must be managed rather than avoided.

ASI will confer a decisive strategic advantage

A decisive strategic advantage (DSA) would mean one actor gains sufficient power to unilaterally determine global outcomes, including the ability to prevent any other actor from challenging their

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[80] Schmid et al., "Evaluating Natural Monopoly Conditions in the AI Foundation Model Market."
[81] Pilz et al., "The US hosts the majority of GPU cluster performance, followed by China."
[82] Kahl and Mitre, "The Real AI Race: America Needs More Than Innovation to Compete With China."
[83] Khan et al., "Should the US Sell Hopper Chips to China?"
[84] Neufeld and Milliken, "Most of America's Top AI Companies Were Founded by Immigrants."
[85] Defending against sophisticated state attackers mounting a determined effort requires extraordinary efforts that may or may not be achievable, see Nevo et al., "Securing AI Model Weights."
[86] Ord, "Inference Scaling Reshapes AI Governance."

position.[87] It is unclear whether ASI would translate into the real-world hard power that might be needed for achieving a DSA.

Reasons for:

- Humans rule the world because of our collective intelligence. Intelligence is very broadly useful across domains: making money, doing science, inventing technologies, campaigning electorally, strategizing geopolitically, etc. So, over time, an actor with sole access to ASI may be able to leverage this advantage to dominate the world.[88]
- If ASI arose rapidly via an intelligence recursion, the leader may have a significant advantage in AI capabilities over the nearest competitor.

Reasons against:

- Other actors may steal, copy, or independently develop ASI, so the leading actor may not have a large lead for long.
- Other great powers may preemptively attack a country that just has or is about to achieve ASI, preventing a DSA from being acquired.[89]
- It is contentious whether ASI would be sufficient to undermine nuclear deterrence.[90] If not, then a country with ASI would still be somewhat constrained by foreign nuclear powers.

Implications for strategic visions:

- If the first actor with ASI is likely to achieve a DSA, this makes it more likely that a single actor, be it a company or government, will have near-total control. Conversely, if a DSA is difficult, stable multilateral solutions are more likely.

The U.S. government will see AI as a top strategic priority

This crux concerns whether the U.S. government will come to view advanced AI as a top priority justifying extraordinary policy measures, akin to how nuclear weapons were seen during the Cold War.

Reasons for:

- The Biden and Trump administrations both foregrounded AI policy, and this growing bipartisan consensus on the central importance of AI seems likely to accelerate as AI capabilities continue to advance rapidly.[91] This might especially be the case if AI becomes

[87] Guest and Delaney, "AI, the spectre of decisive advantage, and international conflict."
[88] Aschenbrenner, "Situational Awareness", Section IIId: The Free World Must Prevail.
[89] Hendrycks et al., "Superintelligence Strategy."
[90] Winter-Levy and Lalwani, "The End of Mutual Assured Destruction?: What AI Will Mean for Nuclear Deterrence."
[91] Friedler and Selbst, "5 points of bipartisan agreement on how to regulate AI."

more tightly integrated into the military and intelligence community, such that AI dominance is seen as strongly linked to broader strategic dominance.

Reasons against:

- Governments have many competing priorities, and bigger picture issues like the geopolitical implications of AI can be neglected.

Implications for strategic visions:

- The stronger the U.S. government's focus is on AI, the more likely it is that we move from the status quo to more strongly differing strategic visions, such as involving stronger domestic regulation or more direct government involvement.

International cooperation is feasible and sustainable

This crux concerns whether meaningful international agreements on AI governance—particularly between rivals, such as the U.S. and China—can be negotiated, implemented, verified, and maintained over time, despite the tense relationship and verification challenges.

Reasons for:

- International cooperation has been moderately successful in curbing global challenges previously, such as minimizing nuclear proliferation and reducing weapons stockpiles, and responding to the hole in the ozone layer.[92]
- Economic interdependence between the U.S. and China makes conflict costly and incentivizes cooperation.

Reasons against:

- AI agreements may be hard to verify without invasive (and perhaps unacceptable) measures, such as allowing international inspectors access to a country's data centers.[93]
- The U.S., particularly under the Trump Administration, is skeptical of international governance and cooperation.[94] This is especially consequential if AI timelines are short, such that the current political and geopolitical order remains in place.
- If AI development has winner-takes-all dynamics, cooperation between competing projects in different countries is harder to sustain because benefits would flow disproportionately to

[92] Knopf, "International Cooperation on WMD Nonproliferation."; UN, "Ozone layer recovery is on track, due to success of Montreal Protocol."
[93] Baker et al., "Verifying International Agreements on AI: Six Layers of Verification for Rules on Large-Scale AI Development and Deployment."
[94] E.g., Michael Kratsios, President Trump's chief science and technology policy advisor, said at the UN: "We totally reject all efforts by international bodies to assert centralized control and global governance of AI." [Kratsios, "Remarks at the Security Council's Open Debate on Artificial Intelligence and International Peace and Security."]

whichever project pulls ahead. This may push countries toward racing rather than coordination.[95]
- International agreements generally take several years to draft, negotiate, and ratify, which may be too slow in short-timelines or fast-takeoff worlds.[96]

Implications for strategic visions:
- International cooperation being more feasible makes the global regulator and joint global project visions more likely.

Governments are more trustworthy AI developers

This crux concerns whether democratic governments or private companies are better positioned to develop advanced AI responsibly. Relevant considerations might include democratic accountability, technical competence, incentive structures, and vulnerability to capture by narrow interests.

Reasons for:
- Democratic governments are backed by the will of the people, providing legitimacy for making high-stakes decisions about the future of AI, and giving them stronger incentives to make decisions that benefit society as a whole.

Reasons against:
- Even if governments are well-intentioned, they may lack the technical expertise and competence to translate intentions into sound policy and practice. So, governments playing a larger role may inadvertently lead to worse outcomes.
- Even if, in theory, governments are democratically accountable, special interest groups may pursue regulatory capture strategies that lead to government policy being unmoored from the public interest, especially if the public is underinformed on an issue.

Implications for strategic visions:
- If private companies are not to be trusted with developing ASI, government-led visions become especially valuable.

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[95] This concern applies to coordination between separate competing projects. Joint international projects sidestep the problem, since participants share in a single outcome rather than each depending on their own project's relative success.
[96] Guest, “Prospects for AI safety agreements between countries.” Sec. “Outside View”

Alignment and control of superintelligence is profoundly difficult

This crux concerns how difficult it will be to ensure that ASI systems reliably pursue goals that their operators and designers intend. A key question is whether alignment and control challenges are tractable engineering problems that can be solved with a manageable level of resources.

Reasons for:

- AI alignment remains an unsolved technical challenge, and there is no reliable roadmap towards a full solution.[97]
- There are strong incentives to race towards ASI, given the potential benefits, but even a single actor building an unaligned ASI may be unrecoverable.[98] This makes the problem more challenging because all ASI projects would need to succeed at alignment, not just some.

Reasons against:

- At current AI capability levels, alignment and control techniques appear to work fairly well.[99]
- Advanced AI agents could do automated alignment research, while themselves being supervised with AI control techniques, allowing us to get AIs to solve key parts of the alignment problem for us.[100]

Implications for strategic visions:

- The stronger one's focus on AI takeover as the most important and difficult risk, the more attractive centralized AI development seems, since this allows one project to invest greatly in alignment and control efforts without worrying about falling behind competitors. Conversely, if AI takeover is relatively easy to prevent, or unlikely in the first place, then pluralistic, multilateral options to distribute power between different human institutions seem more valuable.

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[97] E.g., the International AI Safety Report writes that "Current training methods show progress on mitigating safety hazards from malfunctions and malicious use but remain fundamentally limited." [Bengio et al., "International AI Safety Report 2025," p.191.] See also Hendrycks et al., "Unsolved Problems in ML Safety."
[98] Carlsmith, "Is Power-Seeking AI an Existential Risk?"
[99] E.g., Anthropic has found that their safety training is working increasingly well, with more recent models such as Claude Opus 4.5 scoring better than previous models on safety metrics. [Anthropic, "System Card: Claude Opus 4.5," pp.64-66.]
[100] This is the plan of several frontier AI companies, see e.g., OpenAI's write up here, which is thought to still be the plan despite personnel changes leading to this team no longer existing. [Leike and Sutskever, "Introducing Superalignment."]

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Bibliography

Abraham, Lisa, Joshua Kavner, and Alvin Moon. "A Prisoner's Dilemma in the Race to Artificial General Intelligence." *RAND,* December 2025. https://www.rand.org/pubs/research_reports/RRA4245-1.html.

AI Alignment Forum. "Alignment Tax." 2024. https://www.alignmentforum.org/w/alignment-tax.

Altman, Sam, Greg Brockman, and Ilya Sutskever. "Governance of Superintelligence." *OpenAI*, May 2023. https://openai.com/index/governance-of-superintelligence/.

Amodei, Dario and Matt Pottinger. "Trump Can Keep America's AI Advantage." *The Wall Street Journal*, January 2025. https://www.wsj.com/opinion/trump-can-keep-americas-ai-advantage-china-chips-data-eccdce91.

Amodei, Dario. "The Adolescence of Technology." January 2026. https://www.darioamodei.com/essay/the-adolescence-of-technology.

Anderson-Samways, Bill. "The US Government's Role in Advanced AI Development: Predictions and Scenarios." *Institute for AI Policy and Strategy*, May 2025. https://www.iaps.ai/research/us-government-role-in-advanced-ai-development.

Andreessen, Marc. "Why AI Will Save the World." *Andreessen Horowitz*, June 2023. https://a16z.com/ai-will-save-the-world/.

Andrews, Ivo, Sasha Cadariu, Avital Morris, and David Lambert. "Collective Action Problems." *AI Safety, Ethics, and Society*, 2024. https://www.aisafetybook.com/textbook/collective-action-problems.

Anthropic. "System Card: Claude Opus 4.5." November 2025. https://assets.anthropic.com/m/64823ba7485345a7/Claude-Opus-4-5-System-Card.pdf.

– – –. "The Need for Transparency in Frontier AI." July 2025. https://www.anthropic.com/news/the-need-for-transparency-in-frontier-ai.

Apollo Research. "Internal Deployment of AI Models and Systems in the EU AI Act." August 2025. https://www.apolloresearch.ai/research/internal-deployment-eu-ai-act/.

Aschenbrenner, Leopold. "Situational Awareness: The Decade Ahead." June 2024. https://situational-awareness.ai/.

Askell, Amanda, Miles Brundage, and Gillian Hadfield. "The Role of Cooperation in Responsible AI Development." *arXiv*, July 2019. https://arxiv.org/abs/1907.04534.

Bai, Yuntao, Andy Jones, Kamal Ndousse, et al. "Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback." *arXiv*, April 2022. https://arxiv.org/abs/2204.05862.

Baker, Mauricio, Gabriel Kulp, Oliver Marks, Miles Brundage, and Lennart Heim. "Verifying International Agreements on AI: Six Layers of Verification for Rules on Large-Scale AI Development and Deployment." *arXiv*, July 2025. https://arxiv.org/abs/2507.15916.

Bales, Adam, William D'Alessandro, and Cameron Domenico Kirk-Giannini. "Artificial Intelligence: Arguments for Catastrophic Risk." *Philosophy Compass*, February 2024. https://doi.org/10.1111/phc3.12964.

Belfield, Haydn. "Domestic frontier AI regulation, an IAEA for AI, an NPT for AI, and a US-led Allied Public-Private Partnership for AI: Four institutions for governing and developing frontier AI." *Leverhulme Centre for the Future of Intelligence,* July 2025. https://arxiv.org/pdf/2507.06379.

Bengio, Yoshua, Daniel Privitera, Tamay Besiroglu, et al. "International AI Safety Report 2025." *International AI Safety Report*, January 2025. https://internationalaisafetyreport.org/publication/international-ai-safety-report-2025.

Bostrom, Nick. "Open Global Investment as a Governance Model for AGI." 2025. https://nickbostrom.com/ogimodel.pdf.

Carlsmith, Joe. "AI for AI Safety." *Joe Carlsmith*, March 2025. https://joecarlsmith.com/2025/03/14/ai-for-ai-safety.

– – –. "Is Power-Seeking AI an Existential Risk?" *arXiv*, August 2024. https://arxiv.org/abs/2206.13353.

Chase, Michael S. and William Marcellino. "Incentives for U.S.-China Conflict, Competition, and Cooperation Across Artificial General Intelligence's Five Hard National Security Problems." *RAND,* August 2025. https://www.rand.org/pubs/perspectives/PEA4189-1.html.

Chessen, Matt and Craig Martell. “Beyond a Manhattan Project for Artificial General Intelligence.” *RAND,* April 2025. https://www.rand.org/pubs/commentary/2025/04/beyond-a-manhattan-project-for-artificial-general-intelligence.html.

Clare, Stephen, Sam Manning, and Claire Dennis. “Options and Motivations for International AI Benefit Sharing.” *Centre for the Governance of AI*, February 2025. https://www.governance.ai/analysis/options-and-motivations-for-international-ai-benefit-sharing.

Clymer, Joshua, Nick Gabrieli, David Krueger, and Thomas Larsen. “Safety Cases: How to Justify the Safety of Advanced AI Systems.” *arXiv*, March 2024. https://arxiv.org/abs/2403.10462.

Coalition for a Baruch Plan for AI. “Why a Baruch Plan for AI?” 2024. https://www.cbpai.org/why-a-baruch-plan-for-ai.

Csernatoni, Raluca. “Can Democracy Survive the Disruptive Power of AI?” *Carnegie Endowment For International Peace*, December 2024. https://carnegieendowment.org/research/2024/12/can-democracy-survive-the-disruptive-power-of-ai?lang=en.

Davidson, Tom, Rose Hadshar, and William MacAskill. “Three Types of Intelligence Explosion.” *Forethought*, March 2025. https://www.forethought.org/research/three-types-of-intelligence-explosion.

Delaney, Oscar. “Crucial Considerations in ASI Deterrence.” *Institute for AI Policy and Strategy*, December 2025. https://www.iaps.ai/research/crucial-considerations-in-asi-deterrence.

Delaney, Oscar and Oliver Guest. “Practical Provisions for U.S.-China Cooperation on AI Risks.” *AGI Strategy*, November 2025. https://oscardelaney.substack.com/p/practical-provisions-for-us-china.

Douthat, Ross. “JD Vance on His Faith and Trump’s Most Controversial Policies.” *The New York Times*, May 2025. https://www.nytimes.com/2025/05/21/opinion/jd-vance-pope-trump-immigration.html.

Dung, Leonard. “The Argument for Near-Term Human Disempowerment through AI.” *AI & Society*, vol. 40 (March 2025): 1195–208. https://doi.org/10.1007/s00146-024-01930-2.

Emberson, Luke. “Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023.” *Epoch AI*, January 2026. https://epoch.ai/data-insights/us-vs-china-eci.

Emery-Xu, Nicholas, Andrew Park, and Robert Trager. “Uncertainty, Information, and Risk in International Technology Races.” *AI Governance Initiative*, November 2023.

https://aigi.ox.ac.uk/publications/uncertainty-information-and-risk-in-international-technology-races/.

Emery-Xu, Nicholas, Richard Jordan, and Robert Trager. "International Governance of Advancing Artificial Intelligence." *AI & Society*, vol. 40 (September 2024): 3019–44. https://doi.org/10.1007/s00146-024-02050-7.

Epoch AI. "AI Benchmarking." 2025. https://epoch.ai/benchmarks.

Erdil, Ege, Andrei Potlogea, Tamay Besiroglu, et al. "GATE: An Integrated Assessment Model for AI Automation." *arXiv*, March 2025. https://arxiv.org/abs/2503.04941.

Eth, Daniel and Tom Davidson. "Will AI R&D Automation Cause a Software Intelligence Explosion?" *Forethought*, March 2025. https://www.forethought.org/research/will-ai-r-and-d-automation-cause-a-software-intelligence-explosion.

Felstead, Nicholas. "Enabling Frontier Lab Collaboration to Mitigate AI Safety Risks." *arXiv*, November 2025. https://arxiv.org/pdf/2511.08631.

Finnveden, Lukas. "What's Important in 'AI for Epistemics'?" August 2024. https://lukasfinnveden.substack.com/p/whats-important-in-ai-for-epistemics.

Friedler, Sorelle and Andrew D. Selbst. "5 Points of Bipartisan Agreement on How to Regulate AI." *Brookings*, August 2025. https://www.brookings.edu/articles/five-points-of-bipartisan-agreement-on-how-to-regulate-ai/.

Guest, Oliver. "Prospects for AI Safety Agreements between Countries." *Rethink Priorities*, April 2023. https://rethinkpriorities.org/research-area/prospects-for-ai-safety-agreements-between-countries.

Guest, Oliver and Maria Kostylew. "Undue Influence in an International AI Regulator." *Human General Intelligence*, January 2026. https://humangeneralintelligence.substack.com/p/undue-influence-in-an-international.

Guest, Oliver and Oscar Delaney. "AI, the Spectre of Decisive Advantage, and International Conflict." *Human General Intelligence*, April 2025. https://humangeneralintelligence.substack.com/p/ai-the-spectre-of-decisive-advantage.

Guest, Oliver and Zoe Williams. "Topics for Track IIs: What Can Be Discussed in Dialogues About Advanced AI Risks Without Leaking Sensitive Information?" *Institute for AI Policy and Strategy*, May 2024. https://www.iaps.ai/research/dialogue-topics.

Hadshar, Rose. “Extreme Power Concentration.” *80,000 Hours*, October 2025. https://80000hours.org/problem-profiles/extreme-power-concentration/.

Hao, Karen. *Empire of AI: Dreams and Nightmares in Sam Altman’s OpenAI*. Penguin Random House, 2025.

Hass, Ryan and Colin Kahl. “Laying the Groundwork for US-China AI Dialogue.” *Brookings*, April 2024. https://www.brookings.edu/articles/laying-the-groundwork-for-us-china-ai-dialogue/.

Hausenloy, Jason, Andrea Miotti, and Claire Dennis. “Multinational AGI Consortium (MAGIC): A Proposal for International Coordination on AI.” *arXiv*, October 2023. https://arxiv.org/abs/2310.09217.

Hendrycks, Dan, Eric Schmidt, and Alexandr Wang. “Superintelligence Strategy.” *arXiv*, March 2025. https://arxiv.org/abs/2503.05628.

Hendrycks, Dan, Nicholas Carlini, John Schulman, and Jacob Steinhardt. “Unsolved Problems in ML Safety.” *arXiv*, June 2022. https://arxiv.org/abs/2109.13916.

Hogarth, Ian. “We Must Slow down the Race to God-like AI.” *Financial Times*, April 2023. https://www.ft.com/content/03895dc4-a3b7-481e-95cc-336a524f2ac2.

Kahl, Colin H. and Jim Mitre. “The Real AI Race: America Needs More Than Innovation to Compete With China.” *Foreign Affairs*, July 2025. https://www.foreignaffairs.com/united-states/china-real-artificial-intelligence-race-innovation.

Katzke, Corin and Gideon Futerman. “The Manhattan Trap: Why a Race to Artificial Superintelligence Is Self-Defeating.” *Elsevier*, December 2024. https://arxiv.org/abs/2501.14749.

Khan, Saif, Tao Burga, Tim Fist, and Georgia Adamson. “Should the US Sell Hopper Chips to China?” *Institute for Progress*, December 2025. https://ifp.org/should-the-us-sell-hopper-chips-to-china/.

Knopf, Jeffrey W. *International Cooperation on WMD Nonproliferation*. University of Georgia Press, 2016.

Kohler, Kevin. “CERN for AI: One Analogy, Many Visions.” *Simon Institute for Longterm Governance*, July 2025. https://simoninstitute.ch/blog/post/cern-for-ai-one-analogy-many-visions.

Kratsios, Michael. “Remarks at the Security Council’s Open Debate on Artificial Intelligence and International Peace and Security.” *United States Mission to the United Nations*, September 2025.

https://usun.usmission.gov/remarks-at-the-security-councils-open-debate-on-artificial-intelligence-and-international-peace-and-security/.

Kwa, Thomas, Ben West, Joel Becker, et al. "Measuring AI Ability to Complete Long Tasks." *METR*, March 2025. https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/.

LaForge, Gordon, Allison Stanger, Sarah Myers West, Bruce Schneier, Stephanie Forrest, and Nazli Choucri. "Power and Governance in the Age of AI." *New America*, March 2024. https://www.newamerica.org/planetary-politics/briefs/power-governance-ai-public-good/.

Leike, Jan. "Distinguishing Three Alignment Taxes." *Musings on the Alignment Problem*, December 2022. https://aligned.substack.com/p/three-alignment-taxes.

Leike, Jan and Ilya Sutskever. "Introducing Superalignment." *OpenAI*, July 2023. https://openai.com/index/introducing-superalignment/.

Maas, Matthijs. "Advanced AI Governance: A Literature Review of Problems, Options, and Proposals." *Institute for Law & AI*, November 2023. https://law-ai.org/advanced-ai-gov-litrev/.

Maas, Matthijs and José Villalobos. "International AI Institutions: A Literature Review of Models, Examples, and Proposals." *Institute for Law & AI*, September 2023. https://law-ai.org/international-ai-institutions/.

MacAskill, William. "Better Futures." *Forethought*, August 2025. https://www.forethought.org/research/better-futures.

MacAskill, William and Rose Hadshar. "Intelsat as a Model for International AGI Governance." *Forethought*, March 2025. https://www.forethought.org/research/intelsat-as-a-model-for-international-agi-governance.

Marsh, Nick. "Early US Policy Priorities for AGI." *AI Futures Project*, December 2025. https://blog.ai-futures.org/p/early-us-policy-priorities-for-agi.

METR. "Common Elements of Frontier AI Safety Policies." 2025. https://metr.org/common-elements.

Miotti, Andrea. "An International Treaty to Implement a Global Compute Cap for Advanced Artificial Intelligence." *arXiv*, September 2025. https://arxiv.org/abs/2311.10748.

Miotti, Andrea, Tolga Bilge, Dave Kasten, and James Newport. "A Narrow Path." *Narrow Path,* 2024. https://www.narrowpath.co/.

Mueller, Karl P. "Heeding the Risks of Geopolitical Instability in a Race to Artificial General Intelligence." *RAND,* July 2025. https://www.rand.org/pubs/perspectives/PEA3691-12.html.

Nadella, Satya, Brad Smith, Marc Andreessen, Andreessen Horowitz, and Ben Horowitz. "AI for Startups." *Microsoft*, November 2024. https://blogs.microsoft.com/on-the-issues/2024/11/01/ai-for-startups/.

Narayanan, Arvind and Sayash Kapoor. "AI as Normal Technology." *AI as Normal Technology*, April 2025. https://www.normaltech.ai/p/ai-as-normal-technology.

Neufeld, Jeremy and Lindsay Milliken. "Most of America's Top AI Companies Were Founded by Immigrants." *Institute for Progress*, April 2025. https://ifp.org/most-of-americas-top-ai-companies-were-founded-by-immigrants/.

Nevo, Sella, Dan Lahav, Ajay Karpur, Yogev Bar-On, Henry Alexander Bradley, and Jeff Alstott. "Securing AI Model Weights." *RAND,* May 2024. https://www.rand.org/pubs/research_reports/RRA2849-1.html.

O'Keefe, Cullen. "Chips for Peace: How the U.S. and Its Allies Can Lead on Safe and Beneficial AI." *Lawfare*, July 2024. https://www.lawfaremedia.org/article/chips-for-peace--how-the-u.s.-and-its-allies-can-lead-on-safe-and-beneficial-ai.

Ord, Toby. "Inference Scaling Reshapes AI Governance." February 2025. https://www.tobyord.com/writing/inference-scaling-reshapes-ai-governance.

Patel, Dwarkesh. "Contra Marc Andreessen on AI." *Dwarkesh Podcast*, June 2023. https://www.dwarkesh.com/p/contra-marc-andreessen-on-ai.

– – –. "Mark Zuckerberg — AI Will Write Most Meta Code in 18 Months." *Dwarkesh Podcast*, April 2025. https://www.dwarkesh.com/p/mark-zuckerberg-2.

– – –. "Why I Don't Think AGI Is Right Around the Corner." *Dwarkesh Podcast*, June 2025. https://www.dwarkesh.com/p/timelines-june-2025.

Patell, Liam, and Oliver Guest. "Demonstrating Restraint." *arXiv.* Forthcoming.

Petropoulos, Alex, Bálint Pataki, Daan Juijn, David Janků, and Max Reddel. "Building CERN for AI." *Centre for Future Generations*, January 2025. https://cfg.eu/building-cern-for-ai/.

Pilz, Konstantin F., Robi Rahman, James Sanders, Luke Emberson, and Lennart Heim. "The US Hosts the Majority of GPU Cluster Performance, Followed by China." *Epoch AI*, June 2025. https://epoch.ai/data-insights/ai-supercomputers-performance-share-by-country.

Pilz, Konstantin, Lennart Heim, and Nicholas Brown. "Increased Compute Efficiency and the Diffusion of AI Capabilities." *arXiv*, February 2024. https://arxiv.org/abs/2311.15377.

Scher, Aaron, David Abecassis, Peter Barnett, and Brian Abeyta. “An International Agreement to Prevent the Premature Creation of Artificial Superintelligence.” *arXiv*, November 2025. https://arxiv.org/html/2511.10783v2.

Schmid, Jon, Tobias Sytsma, and Anton Shenk. “Evaluating Natural Monopoly Conditions in the AI Foundation Model Market.” *RAND,* September 2024. https://www.rand.org/pubs/research_reports/RRA3415-1.html.

Shivakumar, Sujai, Charles Wessner, and Thomas Howell. “Balancing the Ledger: Export Controls on U.S. Chip Technology to China.” *Center for Strategic & International Studies,* 2024. https://www.csis.org/analysis/balancing-ledger-export-controls-us-chip-technology-china.

Siddiqui, Saad, Lujain Ibrahim, Kristy Loke, et al. “Promising Topics for U.S.-China Dialogues on AI Risks and Governance.” *arXiv*, May 2025. https://arxiv.org/abs/2505.07468.

Snyder-Beattie, Andrew. “The Four Pillars: A Hypothesis for Countering Catastrophic Biological Risk.” *Defense in Depth*, October 2025. https://defensesindepth.bio/the-four-pillars-a-hypothesis-for-countering-catastrophic-biological-risk/.

Stix, Charlotte, Annika Hallensleben, Alejandro Ortega, and Matteo Pistillo. “The Loss of Control Playbook: Degrees, Dynamics, and Preparedness.” *arXiv*, December 2025. https://arxiv.org/abs/2511.15846.

Todd, Benjamin. “Shrinking AGI Timelines: A Review of Expert Forecasts.” *80,000 Hours*, March 2025. https://80000hours.org/2025/03/when-do-experts-expect-agi-to-arrive/.

Toner, Helen. “Senate Judiciary Written Testimony - Helen Toner.” Senate Committee on the Judiciary, 2024. https://www.judiciary.senate.gov/imo/media/doc/2024-09-17_pm_-_testimony_-_toner.pdf.

Toner, Helen. “‘Long’ Timelines to Advanced AI Have Gotten Crazy Short.” *Rising Tide*, April 2025. https://helentoner.substack.com/p/long-timelines-to-advanced-ai-have.

United Nations. “Ozone Layer Recovery Is on Track, Due to Success of Montreal Protocol.” January 2023. https://news.un.org/en/story/2023/01/1132277.

U.S.-China Economic and Security Review Commission. “2024 Annual Report to Congress.” November 2024. https://www.uscc.gov/annual-report/2024-annual-report-congress.

Vaintrob, Lizka and Owen Cotton-Barratt. “AI Tools for Existential Security.” *Forethought*, March 2025. https://www.forethought.org/research/ai-tools-for-existential-security.

Vance, J.D. “Remarks by the Vice President at the Artificial Intelligence Action Summit in Paris, France.” *The American Presidency Project*, February 2025.

https://www.presidency.ucsb.edu/documents/remarks-the-vice-president-the-artificial-intelligence-action-summit-paris-france.

Wasil, Akash. “Do We Want an ‘IAEA for AI’?” *Lawfare*, November 2024. https://www.lawfaremedia.org/article/do-we-want-an--iaea-for-ai.

Wellerstein, Alex. “Conant on the Role of the British in the Manhattan Project.” *Restricted Data,* 2012. https://blog.nuclearsecrecy.com/2012/03/21/weekly-document-conant-on-the-role-of-the-british-in-the-manhattan-project/.

Winter-Levy, Sam and Nikita Lalwani. “The End of Mutual Assured Destruction?: What AI Will Mean for Nuclear Deterrence.” *Foreign Affairs*, August 2025. https://www.foreignaffairs.com/united-states/artificial-intelligence-end-mutual-assured-destruction.

Yudkowsky, Eliezer. “Artificial Intelligence as a Positive and Negative Factor in Global Risk.” *Machine Intelligence Research Institute,* 2008. https://intelligence.org/files/AIPosNegFactor.pdf.

Zaidi, Waqar and Allan Dafoe. “International Control of Powerful Technology: Lessons from the Baruch Plan for Nuclear Weapons.” *Centre for the Governance of AI*, March 2021. https://www.governance.ai/research-paper/international-control-of-powerful-technology-lessons-from-the-baruch-plan-for-nuclear-weapons.

Zuckerberg, Mark. “Personal Superintelligence.” *Meta*. July 2025. https://www.meta.com/superintelligence/.